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Formal Methods in Software Engg. - E0 272

Assignment 3 - Refinement & Rodin

1) In relational refinement, a client relies on the ADT's pre-condition to know if an operation is safe to invoke. If this pre-condition is met, the client will accept any valid output dictated by the ADT.

On the other hand, if a client invokes an operation outside its precondition, standard totalisation maps it to $\perp$. This makes it clear to the client that the system has crashed.

If we remove this $\perp$, we will allow any operation invoked outside its pre-condition to non-deterministically return a random output. This means we lose the ability to detect failures.

Let us now construct an argument for the above.

i) Let an ADT be defined such that it has a single operation, t. This operation takes no input and outputs an element from $\{0,1\}$.

ii) Abstract ADT A:

→ Precondition for t : True (always safe to call)

→ Effect: Outputs 0 or 1 (non-deterministically)

→ Trace Language $L(A^{\oplus})$: Because the sequence is always safe, the trace covers all strings of 0's & 1's. Hence:

$$L(A^{\oplus}) = (0+1)^*$$

iii) Abstract ADT B

→ Precondition for t : False (can never be called)

→ Effect: undefined

Because the precondition for invoking $\top$ on B is False, the ⊕ totalisation states that B should return a random value from $\{0,1\}$.

Hence, $L(B^{\oplus}) = (0+1)^*$

Because $L(A^{\oplus}) = L(B^{\oplus})$, $L(B^{\oplus}) \subseteq L(A^{\oplus})$. Based on this, we can state that B refines A.

However, any client that is using A will be 'happy' because they expect non-determinism. If this client substitutes B in place of A, it will invoke an operation with a False precondition, leading to unexpected behaviour or a crash.